

$X_1, X_2, \dots, X_B$ variables aléatoires i.d. : $E\{X_i\} = \mu$
 $\downarrow$ $\text{Var}\{X_i\} = \sigma^2$

$$\text{Corr}(X_i, X_j) = \rho$$

$$\text{Corr}(X_i, X_j) = \frac{\text{Cov}(X_i, X_j)}{\sigma_{X_i} \sigma_{X_j}} =$$

identiquement distribuées
 (mais pas indépendantes :
 pas iid)

$$= \frac{E[(X_i - \mu_{X_i})(X_j - \mu_{X_j})]}{\sigma_{X_i} \sigma_{X_j}} =$$

$$= \frac{E[(X_i - \mu)(X_j - \mu)]}{\sigma^2} = \frac{E(X_i X_j) - \overbrace{\mu E(X_j)}^{\mu} - \overbrace{E(X_i) \mu}^{\mu} + \mu^2}{\sigma^2} =$$

$$= \frac{E(X_i X_j) - \mu^2}{\sigma^2} = \rho \Rightarrow \boxed{E(X_i X_j) = \rho \sigma^2 + \mu^2}$$

Nouvelle variable aléatoire :

$$Y = \frac{1}{B} (X_1 + \dots + X_B)$$

$$E(Y) = \frac{1}{B} [\underbrace{E(X_1)}_{\mu} + \dots + \underbrace{E(X_B)}_{\mu}] = \frac{B\mu}{B} = \mu = \mu_Y$$

$$\text{Var}(Y) = ?$$

$$\text{Var}(Y) = E[(Y - \mu_Y)^2] = E(Y^2 - 2\mu_Y Y + \mu_Y^2) =$$

$$= E(Y^2) - 2\mu_Y E(Y) + \mu_Y^2 =$$

$$= E(Y^2) - \mu_Y^2$$

$$= E(Y^2) - \mu^2$$

v.a. X $\mu_X = \mu$ $\sigma_X = \sigma$

$$Y = kX$$

$$\mu_Y = E[kX] = kE[X] = k\mu$$

$$\sigma_Y^2 = E[(Y - \mu_Y)^2] = E[Y^2] - \mu_Y^2 =$$

$$= E[k^2 X^2] - k^2 \mu^2 = k^2 E[X^2] - k^2 \mu^2 \Rightarrow$$

$$\sigma_Y^2 = k^2 (E\{x^2\} - \mu^2) = k^2 \sigma_X^2$$

$$\boxed{\sigma_Y^2 = k^2 \sigma_X^2}$$

$$Y = \frac{1}{B} (X_1 + \dots + X_B) \quad \mu_Y = \mu$$

$$\text{Var}(Y) = E(Y^2) - \mu_Y^2 = E(Y^2) - \mu^2 =$$

$$= E \left[\left(\frac{X_1 + \dots + X_B}{B} \right)^2 \right] - \mu^2 = \frac{1}{B^2} E[(X_1 + \dots + X_B)^2] - \mu^2 =$$

$$= \frac{1}{B^2} \left[E(X_1^2 + \dots + X_B^2 + 2 \sum_{i \neq j} X_i X_j) \right] - \mu^2 =$$

$$= \frac{1}{B^2} \left(E(X_1^2) + \dots + E(X_B^2) + \sum_{i \neq j} E(X_i X_j) \right) - \mu^2 =$$

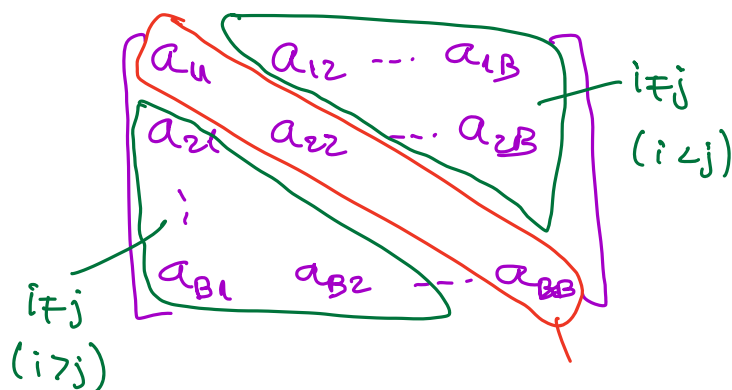
$$\text{Var}(X_i) = \sigma_{X_i}^2 = E(X_i^2) - \mu_{X_i}^2 \Rightarrow E(X_i^2) = \mu^2 + \sigma^2, \quad i=1 \dots B$$

$$= \frac{1}{B^2} (B(\mu^2 + \sigma^2) + (\dots)) - \mu^2$$

$$\boxed{E(X_i X_j) = \sigma^2 + \mu^2}$$

Combien de termes dans
la somme $\sum_{i \neq j} \dots$?

$$2 \times \frac{B(B-1)}{2} = B(B-1)$$



$$\frac{B^2 - B}{2} = \frac{B(B-1)}{2}$$

termes

$i = j$
B termes

$$\text{Var}(Y) = \frac{1}{B^2} \left(B(\mu^2 + \sigma^2) + \frac{B(B-1)}{(B^2 - B)} (\sigma^2 + \mu^2) \right) - \mu^2$$

$$= \frac{1}{B^2} (\cancel{B\mu^2} + \underline{B\sigma^2} + B^2\beta\sigma^2 + \cancel{B^2\mu^2} - \underline{B\beta\sigma^2} - \cancel{B\mu^2}) - \cancel{\mu^2}$$

$$= \frac{1}{B^2} (B^2\beta\sigma^2 + B(1-\beta)\sigma^2)$$

$$\boxed{\text{Var}(Y) = \beta\sigma^2 + \frac{1-\beta}{B}\sigma^2}$$